

Math 125 Final Review 2

Solutions on the following page.

1. Fall 2019 Q2 b-c
2. Fall 2018 Q3 c-d
3. Fall 2017 Q11

1. Q1

(a) $\tan \theta = y/x$

(b) $\theta = \pi/3$

(c) Use chain rule to find $d\theta/dt = 7/8$

2. Q2

(a)

$$3u^{4/3} = 3(1 + x^{1/4})^{4/3} + C$$

(b) 0 by oddness

3. Q3

(a) $g'(x) = \frac{1}{t^2-t+1}$

(b) $g''(x) = \frac{1-2t}{(t^2-t+1)^2}$

(c) $x < 1/2$